



Complete Genome Sequence of Bacteriophage Fizzles, Isolated from *Microbacterium foliorum*

Skyler Adams,^a Gabrielle Spatz,^a Riley Babcock,^a Chloe Butler,^a Samantha Conger,^a Madison Crew,^a Stephanie Garcia,^a Joanna Gonzalez,^a Jocelyn Hodges,^a Alondra Martinez,^a Samuel Munoz,^a Chloe O'Grady,^a Abigail Quirl,^a Kristin Sefcik,^a Tanner Taylor,^a Gustavo Vazquez,^a Faith Cox,^a  Dustin Edwards^a

^aDepartment of Biological Sciences, Tarleton State University, Stephenville, Texas, USA

ABSTRACT Microbacteriophage Fizzles has a 62,078-bp linear double-stranded DNA genome sequence, predicted to contain 104 protein-coding genes. Fizzles is a *Siphoviridae* actinobacteriophage isolated from an ant hill soil sample collected in Stephenville, TX. Microbacteriophage Fizzles has >83.6% nucleotide identity with microbacteriophages Squash and Nike.

To further understand bacteriophage genomic diversity as part of the Howard Hughes Medical Institute Science Education Alliance Phage Hunters Advancing Genomic and Evolutionary Science program (1), we report the genome sequence of microbacteriophage Fizzles (2). Fizzles was extracted from dry soil collected from an ant bed in Stephenville, TX, USA (global positioning system [GPS] coordinates, 32.2197N, 98.1989W). The soil samples were washed with peptone-yeast extract-calcium (PYCa) liquid medium, and bacteriophages were extracted through a 0.22- μ m filter. The filtered medium was mixed with soft agar containing *Microbacterium foliorum* strain SEA B-24224, overlaid on PYCa agar, and incubated at 29°C for 48 h. Bacteriophage replication produced small, lytic plaques. Fizzles was isolated by two rounds of picking a single, well-separated plaque, followed by diluting the bacteriophage sample in a 10-fold dilution series and plating with *M. foliorum*. Negative-staining transmission electron microscopy showed *Siphoviridae* morphology, with a tail length of 160 nm and an isometric capsid 65 nm in diameter (Fig. 1), as measured using ImageJ v1.53m (3).

High-titer lysate was produced by flooding "webbed" plates, as described in the *Phage Discovery Guide* (4). Genomic DNA was extracted from high-titer lysate using a modified zinc chloride precipitation method (4, 5). Genomic sequencing libraries were prepared using the NEBNext Ultra II kit (New England Biolabs, Ipswich, MA) and sequenced using an Illumina MiSeq instrument at the Pittsburgh Bacteriophage Institute (Pittsburgh, PA). Sequencing was performed to 2,719-fold coverage from 1,194,881 total reads of 150-bp read length. Assembly to produce a single-bacteriophage contig was performed using Newbler v2.9 with default settings, and Consed v2.9.0 was used for assembly of a single bacteriophage contig to ensure performance for quality control (6, 7). Fizzles has a double-stranded linear DNA genome 62,078 bp long, with 181-bp direct terminal repeats and a 68.2% G+C content. Whole-genome nucleotide alignment using BLASTn (<https://blast.ncbi.nlm.nih.gov/>) (8) showed that Fizzles has a nucleotide sequence identity of >83.6% with *Microbacterium* phages Squash (GenBank accession number [MH153813](#)) and Nike ([MT114166](#)).

Initial auto-annotation was performed using GLIMMER v3.02 (9) and GeneMark v2.5p (10, 11), and the annotation was manually refined using Phamerator (<http://phamerator.org/>), DNA Master v5.23.2 (<http://phagesdb.org/DNAMaster/>), and PECAAN. No tRNA genes were identified using Aragorn v1.2.38 (12) and tRNAscan-SE v2.0 (13). Putative functions for 72 of 104 predicted protein-coding genes were assigned using NCBI BLASTp (8) and HHpred v3.0beta (14, 15). All tools were run with default parameters. Start codon

Editor John J. Dennehy, Queens College CUNY

Copyright © 2022 Adams et al. This is an open-access article distributed under the terms of the [Creative Commons Attribution 4.0 International license](#).

Address correspondence to Dustin Edwards, dc Edwards@tarleton.edu.

The authors declare no conflict of interest.

Received 3 November 2021

Accepted 6 December 2021

Published 6 January 2022

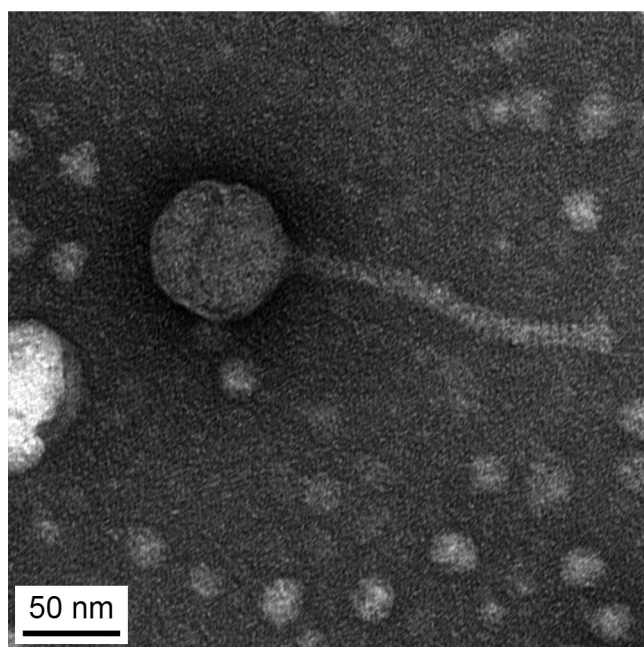


FIG 1 Transmission electron microscopy (TEM) of microbacteriophage Fizzles. High-titer lysate was placed on a carbon type-B 300-mesh grid, stained with 1% uranyl acetate, and imaged using an FEI Tecnai G² Spirit BioTwin instrument (NL1.160G). TEM micrographs of negatively stained Fizzles particles show a tail length of 160 nm and an icosahedral capsid 65 nm in diameter.

usage was 77.7% for AUG, 21.4% for GUG, and 0.97% for UUG. Genes were transcribed rightward (48.1% of genome) and leftward (51.9% of genome) and encode structural proteins, histidine triad nucleotide-binding protein, hydrolase, MazG-like nucleotide pyrophosphohydrolase, HNH endonuclease, RuvC-like resolvase, DNA primase/helicase, RecA-like DNA recombinase, nucleotide pyrophosphohydrolase, and DnaJ-like chaperonin.

Data availability. The genome sequence for microbacteriophage Fizzles is available at GenBank under accession number [MW924638](#). The raw reads are available under SRA accession number [SRX11067172](#).

ACKNOWLEDGMENTS

Support for this research was provided by Tarleton State University College of Science and Technology and by the Howard Hughes Medical Institute Science Education Alliance-Phage Hunters Advancing Genomics and Evolutionary Science (SEA-PHAGES) research and education program.

We thank Graham Hatfull, Welkin Pope, Deborah Jacobs-Sera, Daniel Russell, Rebecca Garlena, Sally Malloy, and Phoebe Doss for their technical support during the imaging of the virion and the isolation, sequencing, and annotation of this genome.

REFERENCES

- Jordan TC, Burnett SH, Carson S, Caruso SM, Clase K, DeJong RJ, Dennehy JJ, Denver DR, Dunbar D, Elgin SCR, Findley AM, Gissendanner CR, Golebiewska UP, Guild N, Hartzog GA, Grillo WH, Hollowell GP, Hughes LE, Johnson A, King RA, Lewis LO, Li W, Rosenzweig F, Rubin MR, Saha MS, Sandoz J, Shaffer CD, Taylor B, Temple L, Vazquez E, Ware VC, Barker LP, Bradley KW, Jacobs-Sera D, Pope WH, Russell DA, Cresawn SG, Lopatto D, Bailey CP, Hatfull GF. 2014. A broadly implementable research course in phage discovery and genomics for first-year undergraduate students. *mBio* 5:e01051-13. <https://doi.org/10.1128/mBio.01051-13>.
- Russell DA, Hatfull GF. 2017. PhagesDB: the actinobacteriophage database. *Bioinformatics* 33:784–786. <https://doi.org/10.1093/bioinformatics/btw711>.
- Schneider CA, Rasband WS, Eliceiri KW. 2012. NIH Image to ImageJ: 25 years of image analysis. *Nat Methods* 9:671–675. <https://doi.org/10.1038/nmeth.2089>.
- Poxleitner M, Pope W, Jacobs-Sera D, Sivanathan V, Hatfull G. 2018. Phage discovery guide. Howard Hughes Medical Institute, Chevy Chase, MD.
- Santos MA. 1991. An improved method for the small scale preparation of bacteriophage DNA based on phage precipitation by zinc chloride. *Nucleic Acids Res* 19:5442. <https://doi.org/10.1093/nar/19.19.5442>.
- Russell DA. 2018. Sequencing, assembling, and finishing complete bacteriophage genomes. *Methods Mol Biol* 1681:109–125. https://doi.org/10.1007/978-1-4939-7343-9_9.
- Gordon D, Green P. 2013. Consed: a graphical editor for next-generation sequencing. *Bioinformatics* 29:2936–2937. <https://doi.org/10.1093/bioinformatics/btt515>.

8. Altschul SF, Gish W, Miller W, Myers EW, Lipman DJ. 1990. Basic local alignment search tool. *J Mol Biol* 215:403–410. [https://doi.org/10.1016/S0022-2836\(05\)80360-2](https://doi.org/10.1016/S0022-2836(05)80360-2).
9. Delcher AL, Harmon D, Kasif S, White O, Salzberg SL. 1999. Improved microbial gene identification with GLIMMER. *Nucleic Acids Res* 27:4636–4641. <https://doi.org/10.1093/nar/27.23.4636>.
10. Borodovsky M, Mills R, Besemer J, Lomsadze A. 2003. Prokaryotic gene prediction using GeneMark and GeneMark.hmm. *Curr Protoc Bioinformatics* 1:4.5.1–4.5.16. <https://doi.org/10.1002/0471250953.bi0405s01>.
11. Besemer J, Borodovsky M. 2005. GeneMark: Web software for gene finding in prokaryotes, eukaryotes and viruses. *Nucleic Acids Res* 33:W451–W454. <https://doi.org/10.1093/nar/gki487>.
12. Laslett D, Canback B. 2004. ARAGORN, a program to detect tRNA genes and tmRNA genes in nucleotide sequences. *Nucleic Acids Res* 32:11–16. <https://doi.org/10.1093/nar/gkh152>.
13. Lowe TM, Chan PP. 2016. tRNAscan-SE On-line: integrating search and context for analysis of transfer RNA genes. *Nucleic Acids Res* 44:W54–W57. <https://doi.org/10.1093/nar/gkw413>.
14. Söding J, Biegert A, Lupas AN. 2005. The HHpred interactive server for protein homology detection and structure prediction. *Nucleic Acids Res* 33:W244–W248. <https://doi.org/10.1093/nar/gki408>.
15. Zimmermann L, Stephens A, Nam S-Z, Rau D, Kübler J, Lozajic M, Gabler F, Söding J, Lupas AN, Alva V. 2018. A completely reimplemented MPI Bioinformatics Toolkit with a new HHpred server at its core. *J Mol Biol* 430:2237–2243. <https://doi.org/10.1016/j.jmb.2017.12.007>.